New Imbalanced 30K External Evaluation

20,979 legitimate + 9,021 phishing (30.07% phishing)

30,000 unique URLs, hostnames and encoded identities
SHA-256: da61d5976f0db1c59a07101a565a077bb6997d7c8ce87cc2a7e8bf5688e84095

Best balanced configuration: HYB-MS +10%
F1 91.98% | ROC-AUC 96.99% | FPR 0.96%

All checkpoints and thresholds were frozen before this test.
No retraining, recalibration, or test-set threshold tuning.

Complete metrics — overall ranks 1 to 10

Model	Cond.	ACC	PREC	REC	F1	ROC-AUC	PR-AUC	FPR	FNR	Mean rank
HYB-MS	+10%	95.43	97.49	87.05	91.98	96.99	96.38	0.96	12.95	5.8
HYB-MS	Real	94.64	96.51	85.25	90.53	96.94	95.97	1.33	14.75	7.0
CNN+PSL	Real	45.61	35.51	97.29	51.98	92.54	88.18	76.61	2.71	7.0
HYB-MS	+20%	94.23	98.27	82.26	89.56	96.58	95.89	0.62	17.74	8.2
HYB-MS	+30%	92.07	98.47	74.78	85.01	95.92	94.69	0.50	25.22	8.6
CNN+PSL	+40%	49.57	37.47	95.94	53.75	89.97	85.38	70.37	4.06	9.2
CNN+PSL	+30%	44.89	35.68	96.63	51.89	90.73	85.64	77.36	3.37	9.2
HYB-MS	+40%	90.85	98.46	70.69	82.29	95.71	94.31	0.48	29.31	9.4
MS-LSTM	Real	34.16	31.06	97.22	47.07	93.54	92.85	92.95	2.78	10.2
CNN+PSL	+10%	37.07	32.02	97.01	48.14	90.38	86.00	88.71	2.99	10.5

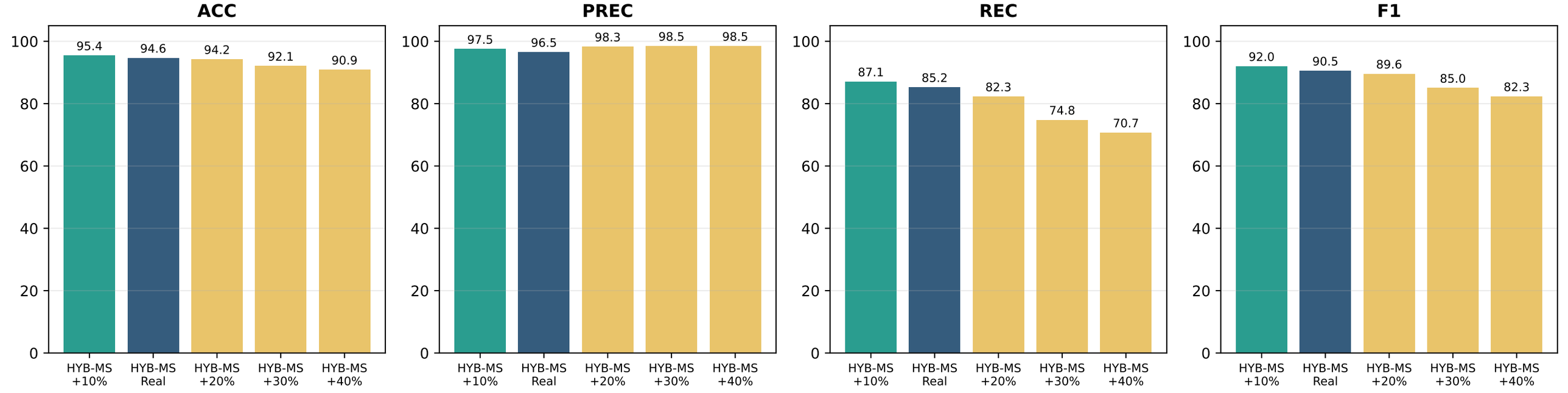
Complete metrics — overall ranks 11 to 20

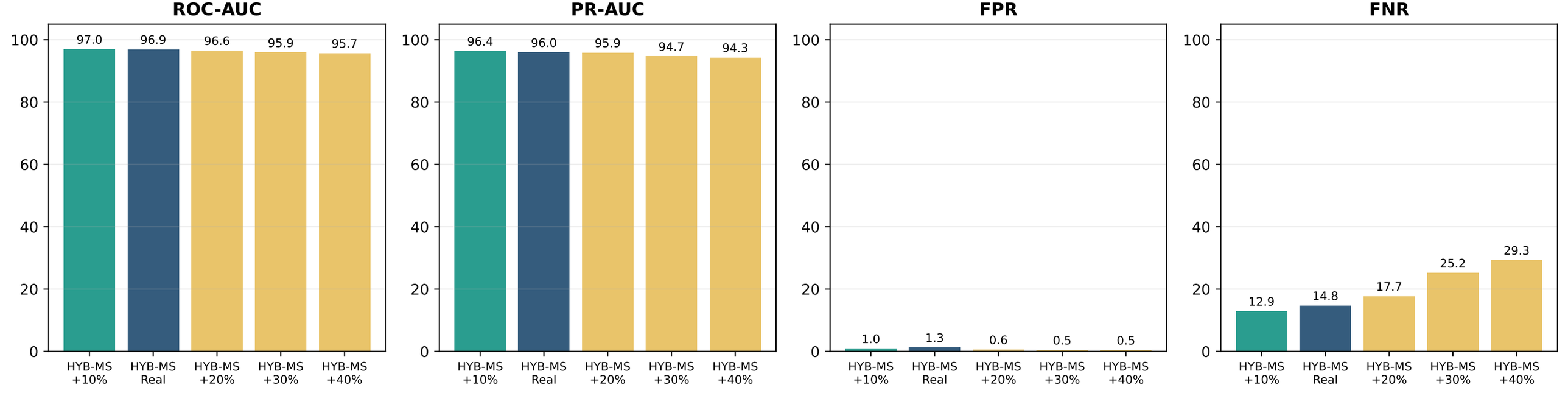
Model	Cond.	ACC	PREC	REC	F1	ROC-AUC	PR-AUC	FPR	FNR	Mean rank
CNN+PSL	+20%	44.81	35.05	96.50	51.38	90.17	85.34	77.42	3.50	10.6
RF-PSL	Real	40.70	33.17	95.82	49.28	89.73	87.58	83.00	4.18	11.9
MS-CNN	Real	30.48	29.86	97.25	45.69	82.01	71.76	98.24	2.75	13.4
RF-PSL	+40%	29.56	29.71	98.32	45.64	48.78	45.94	100.00	1.68	15.0
ORIG-LSTM	Real	30.25	29.72	96.67	45.46	84.11	78.48	98.31	3.33	15.0
MS-RNN	Real	29.44	29.48	96.68	45.18	87.02	81.45	99.47	3.32	16.1
RF-PSL	+30%	29.54	29.69	98.22	45.60	47.99	41.23	100.00	1.78	16.2
RF-PSL	+20%	29.52	29.68	98.17	45.59	46.48	41.75	100.00	1.83	16.5
RF-PSL	+10%	29.40	29.60	97.78	45.44	52.22	46.38	100.00	2.22	16.6
ORIG-CNN	Real	29.24	29.31	95.85	44.89	63.04	46.07	99.40	4.15	19.0

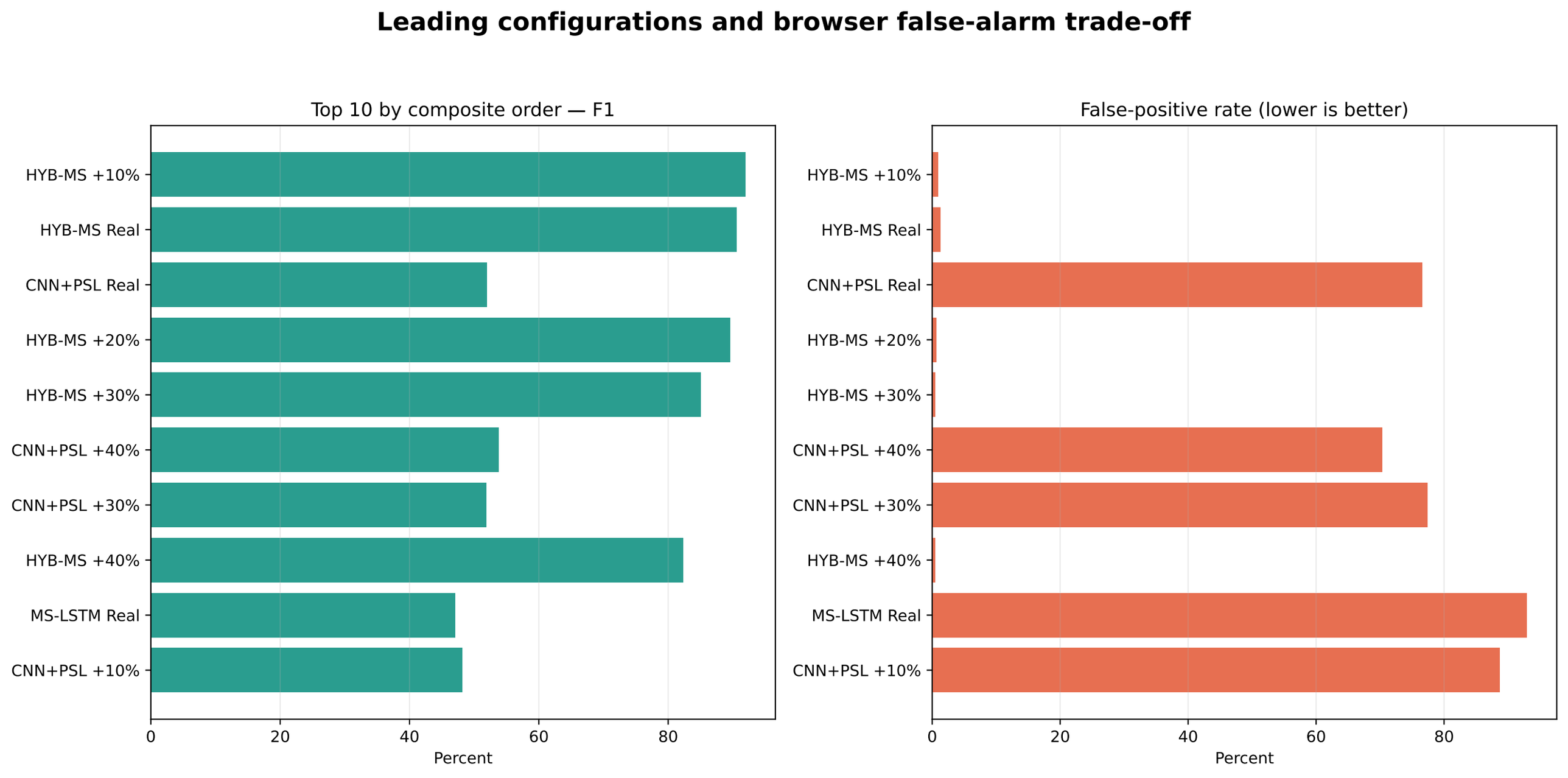
Complete metrics — overall ranks 21 to 30

Model	Cond.	ACC	PREC	REC	F1	ROC-AUC	PR-AUC	FPR	FNR	Mean rank
ORIG-RNN	Real	28.86	29.18	95.69	44.72	74.39	65.07	99.88	4.31	19.6
SVM-PSL	Real	32.00	28.48	83.49	42.48	52.95	45.00	90.14	16.51	20.1
MS-RF	Real	38.01	26.31	58.91	36.37	53.13	51.05	70.98	41.09	20.6
SVM-PSL	+10%	28.74	27.73	85.28	41.85	43.08	36.27	95.57	14.72	21.8
SVM-PSL	+20%	27.40	27.31	85.13	41.36	37.44	31.92	97.43	14.87	23.2
MS-SVM	Real	43.38	18.30	25.50	21.31	30.92	34.55	48.93	74.50	23.2
ORIG-SVM	Real	43.84	17.96	24.31	20.66	29.46	32.86	47.76	75.69	23.8
SVM-PSL	+40%	27.00	27.16	84.90	41.16	31.64	27.43	97.90	15.10	24.5
SVM-PSL	+30%	26.87	27.11	84.80	41.09	33.70	29.09	98.04	15.20	25.0
ORIG-RF	Real	18.99	17.55	45.83	25.39	21.60	25.55	92.55	54.17	27.5

Calibrated HYB-MS ensembles on the frozen imbalanced 30K






Leading configurations and browser false-alarm trade-off
Top 10 by composite order — F1
HYB-MS +10%
HYB-MS Real
CNN+PSL Real
HYB-MS +20%
HYB-MS +30%
CNN+PSL +40%
CNN+PSL +30%
HYB-MS +40%
MS-LSTM Real
CNN+PSL +10%
0
20
40
60
80
Percent
False-positive rate (lower is better)
HYB-MS +10%
HYB-MS Real
CNN+PSL Real
HYB-MS +20%
HYB-MS +30%
CNN+PSL +40%
CNN+PSL +30%
HYB-MS +40%
MS-LSTM Real
CNN+PSL +10%
0
20
40
60
80
Percent

Interpretation

HYB-MS +10% is best overall under the stated equal-weight metric ranking (5.8).
Compared with calibrated real-only, F1 changes 90.53% → 91.98%.
Recall changes 85.25% → 87.05%; FNR changes 14.75% → 12.95%.
FPR changes 1.33% → 0.96%.

The +20% to +40% conditions reduce FPR further, but lose too much phishing recall and F1.
The result confirms the earlier frozen 10K direction without changing the chosen +10% condition.

Limitations
• Tranco roots are legitimate proxies rather than individually audited pages.
• Legitimate roots and phishing full URLs differ structurally.
• PhishTank's current feed contains still-online historical reports, so this is not wholly temporal.
• 30.07% phishing is a seeded stress-test prevalence, not measured browser prevalence.
• Strong metrics support a deployment candidate, not deployment readiness by themselves.